

GEMM pseudocode — one row tile

```
v1 = vsetvl_e32m2(rows)
initialize r[0:8]
for k in unroll(K, 8):
    a = vle32_v_f32m2(Aptr, v1)
    for q in unroll(8):
        r[q] = fma_m2(r[q], B[k,q], a)
    advance A stream
write back r[0:8]
```

Target profile: P1–M1

Integer LMUL = 1

m2 intrinsics excluded

Software-enforced on Spacemit X60

(a) LLM-based migration

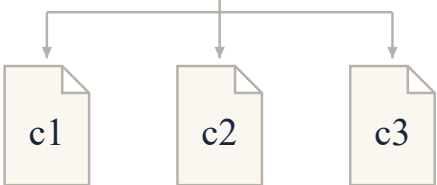
Generated implementation

Source kernel

RVV intrinsics · loops · reuse

Target constraints

P1–M1 · integer LMUL = 1



Generated candidates

Illustrative outputs

(b) GapMigrate (ours)

Hierarchical mapping

$g \in \{8, 4, 2\}$, shown: $g = 4$

```
for each m1 row tile:
    for Q in groups(8, 4):
        reset A stream
        initialize r[Q]
        source reduction(Q, m1)
        write back r[Q]
```

4 live accumulators



(c) Fixed-organization migration

Intrinsic legalization

$g = 8$

```
for each m1 row tile:
    reset A stream
    initialize r[0:8]
    source reduction(0:8, m1)
    write back r[0:8]
```

8 live accumulators

